

Notes: Ben-Rephael, Kandel, and Wohl (JFE, 2012)

Measuring Investor Sentiment with Mutual Fund Flows

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Research question and core hypothesis	3
3.1	Core hypothesis	3
3.2	Key distinction	3
4	Data and variables	3
4.1	Data sources	3
4.2	Equity fund group	3
4.3	Main variables	4
5	Empirical design	4
5.1	Contemporaneous regressions	4
5.2	Predictive regressions	4
5.3	VAR impulse responses	4
5.4	Trading-strategy tests	4
5.5	Risk-premium and sentiment checks	4
5.6	Cross-sectional tests	4
6	Figures and tables	5
6.1	Fund-flow construction and summary statistics	5
6.2	Return reversal and dynamic price response	7
6.3	Trading strategy and robustness	9
6.4	Links to VIX, rates, and sentiment measures	11
6.5	Cross-sectional evidence: small stocks and growth stocks	12
6.6	Appendix VAR relations	13
7	Key mechanisms and interpretation	13
7.1	Why net exchanges matter	13
7.2	Why reversal matters	14
7.3	Why VIX is not enough	14
7.4	Why the effect is stronger for small and growth stocks	14
8	Limitations	14
9	Short summary	14
10	Conclusion	15

1 Big picture

This paper proposes a mutual-fund-flow based measure of **aggregate investor sentiment**: monthly shifts between bond funds and equity funds. The authors focus on **aggregate net exchanges into equity funds**, denoted *NEIO*. These are within-family exchanges into equity funds minus exchanges out of equity funds, normalized by prior fund assets.

Main finding

Monthly net exchanges into equity funds are strongly positively related to contemporaneous stock market returns, but the price effect mostly reverses afterward. A one-standard-deviation shock to *NEIO* is associated with about **1.95%** contemporaneous excess market return; about **85%** of the effect reverses within four months, and essentially all of it reverses within ten months.

- Net exchanges are much more informative than total net flows or net sales/redemptions.
- The reversal pattern is economically meaningful and supports feasible market-timing strategies based on publicly available lagged flow data.
- *NEIO* is negatively related to changes in VIX, but the return reversal is too large to be explained by time-varying risk premia.
- The effect is stronger for smaller stocks and growth stocks, consistent with a sentiment interpretation.

Interpretation: retail mutual fund investors' shifts between bond and equity funds proxy for sentiment-driven demand shocks. These shocks move aggregate market prices temporarily, producing "noise" that later corrects.

2 Incremental contribution of the paper

1. **Separates net exchanges from ordinary fund flows.** Earlier studies emphasize aggregate net equity fund flows. This paper isolates within-family exchanges between bond-like and equity-like fund categories, which are closer to active asset-allocation shifts.
2. **Documents aggregate return reversal.** Prior work finds a contemporaneous flow-return relation but little predictability afterward. This paper shows that net exchanges predict subsequent reversals.
3. **Provides a sentiment measure for the aggregate market.** The paper argues that *NEIO* captures a dimension of sentiment distinct from the University of Michigan consumer sentiment index and Baker–Wurgler sentiment.
4. **Shows economic magnitude.** A one-standard-deviation *NEIO* shock is associated with a 1.95% contemporaneous excess market return, followed by near-complete reversal.
5. **Connects aggregate sentiment to cross-sectional sentiment predictions.** The return effect is stronger among small stocks and growth stocks, which are typically more sentiment-sensitive.
6. **Extends the sample.** The 1984–2008 sample is substantially longer than the samples in Warther, Edwards–Zhang, and Fant.

3 Research question and core hypothesis

The paper asks whether mutual fund flows can reveal investor sentiment that temporarily moves aggregate stock market prices.

3.1 Core hypothesis

If mutual fund investors shift money from bond funds to equity funds for sentiment or non-fundamental reasons, then stock market prices should rise contemporaneously. If the shift is non-fundamental, the price change should reverse over subsequent months.

3.2 Key distinction

The paper distinguishes three flow concepts:

- **Net flows:** new sales plus exchanges in minus redemptions and exchanges out.
- **Net sales/redemptions (NSR):** new sales minus redemptions. These are more related to savings and withdrawals.
- **Net exchanges (NEIO):** exchanges in minus exchanges out. These are more likely to reflect active reallocation between bond and equity funds within fund families.

The paper's key empirical move is to focus on *NEIO*, not total net flows.

4 Data and variables

4.1 Data sources

- **Fund-flow data:** Investment Company Institute monthly aggregate mutual fund category data from February 1984 to December 2008.
- **Market returns:** CRSP value-weighted NYSE/Amex/Nasdaq market return minus the 30-day T-bill return.
- **Risk and sentiment controls:** VIX, short-term interest rates, University of Michigan Consumer Sentiment Index, and Baker–Wurgler sentiment.
- **Cross-sectional portfolios:** Fama–French size and book-to-market portfolios, plus Russell 1000 and Russell 2000 returns.

4.2 Equity fund group

The equity group includes domestic equity funds, international equity funds, and mixed funds. The bond group includes bond funds and money market funds. Because net exchanges sum to zero across all fund categories, net exchanges into equity funds are the mirror image of net exchanges out of bond funds.

4.3 Main variables

- **NEIO:** normalized net exchanges into equity funds.
- **NSR:** normalized net sales minus redemptions.
- **ExRET:** monthly market return minus the 30-day T-bill return.
- **SumNEIO2_4:** sum of *NEIO* from months $t - 2$ through $t - 4$, used because ICI data are released with a one-month lag.

5 Empirical design

5.1 Contemporaneous regressions

The authors first regress monthly excess market returns on contemporaneous flow measures. The key comparison is between total net flows, net sales/redemptions, and net exchanges. The result is that *NEIO* dominates: it explains far more of the contemporaneous return relation than *NSR*.

5.2 Predictive regressions

The authors regress future market excess returns on lagged *NEIO*. Negative coefficients on lagged *NEIO* indicate reversal. They focus especially on *SumNEIO2_4*, because the most recent month of ICI flow data is not publicly available at the start of the return month.

5.3 VAR impulse responses

The paper estimates vector autoregressions using *NSR*, *NEIO*, and excess market returns. The impulse-response analysis quantifies the full dynamic path after a one-standard-deviation *NEIO* shock.

5.4 Trading-strategy tests

The paper constructs feasible strategies using lagged, publicly available *NEIO*. The strategy takes a position opposite to the sentiment signal: if past net exchanges into equity are high, hold T-bills; if past net exchanges are low, hold the stock market.

5.5 Risk-premium and sentiment checks

The paper examines whether *NEIO* is merely capturing changes in risk premia through VIX or interest rates. It also compares *NEIO* to survey and Baker–Wurgler sentiment measures.

5.6 Cross-sectional tests

If *NEIO* measures sentiment, its effects should be stronger where sentiment effects are typically stronger: small stocks and growth stocks. The paper tests this using Fama–French and Russell portfolios.

6 Figures and tables

6.1 Fund-flow construction and summary statistics

Table 1

Mutual fund flows in January 2003.

The table presents the monthly flows of mutual funds in January 2003 (in millions of dollars). ICI data include four flow categories: new sales, redemptions, "exchanges in," and "exchanges out." The funds are classified into four major investment types: bonds, domestic equity (with five sub-investment categories), international equity, and mixed funds (funds that contain both equity and bonds). New sales and redemptions are actual cash flows that enter or exit the fund family. "Exchanges in" and "exchanges out" are transfers between categories in the same fund family. Net flows are the fund flows from the sum of the four categories: "new sales" minus "redemptions" plus "exchanges in" minus "exchanges out."

Fund categories	Exchanges in	Exchanges out	Net exchanges	New sales	Redemptions	Net sales	Net flows
Bonds	36,839	32,814	4,025	1,075,466	1,067,709	7,758	11,783
Aggressive growth	6,421	7,485	-1,064	12,109	13,052	-943	-2,007
Growth	5,027	6,607	-1,580	18,335	17,344	991	-589
Growth and income	3,956	4,754	-798	17,501	15,554	1,946	1,148
Income equity	769	759	10	2,474	1,745	730	739
Sector	2,622	2,943	-321	3,057	3,346	-289	-610
Total domestic equity	18,794	22,547	-3,754	53,477	51,042	2,435	-1,319
International equity	4,325	4,642	-318	18,623	17,249	1,374	1,057
Mixed	1,376	1,333	43	7,150	6,100	1,050	1,093
Total categories	61,333	61,336	-3	1,154,716	1,142,099	12,617	12,613

Table 1. Mutual fund flows in January 2003.

Table 2

Summary statistics.

The table presents the summary statistics of the equity fund flows, asset values, and equity market returns. The sample period ranges from February 1984 to December 2008, covering a total of 299 months. Fund flows and net asset values were obtained from the Investment Company Institute (ICI) and include the following fund categories: domestic equity, international equity, and mixed funds. The total value of the equity market and returns were obtained from the Center for Research in Security Prices (CRSP) as the total value and value-weighted return of NYSE, Amex, and Nasdaq stocks. *NET_FLOWS* is the net flows (in billion \$) into and out of the funds from four components: "new sales" plus "exchanges in" minus "redemptions" minus "exchanges out." Assets are the end-of-period net assets of the equity funds (in billions \$). The net flows of the equity funds and their components are normalized each month by the previous month's fund assets value: *NFLOWS* is the normalized net flows (in %). *NS* is the normalized "new sales" (in %). *NR* is the normalized "redemptions" (in %). *NI* is the normalized "exchanges in" (in %). *NO* is the normalized "exchanges out" (in %). *NSR* is the normalized net sales ("new sales" minus "redemptions") in %. *NEIO* is the normalized "net exchanges" ("exchanges in" minus "exchanges out") in %. *RET* is the monthly market return from the CRSP value-weighted index composed of NYSE, Amex, and Nasdaq stocks (in %). *EXRET* is the excess market return (over 30-day T-bill return) in %.

Variables	Mean	Median	Std	Min	Max
<i>NET_FLOWS</i> (\$ billions)	7.89	7.94	14.34	-86.25	50.35
<i>ASSETS</i> (\$ billions)	2,405.4	1,737.1	2,201.4	79.0	7,675.9
<i>NFLOWS</i> (%)	0.61	0.51	0.87	-2.94	3.85
<i>NS</i> (%)	2.29	2.21	0.59	0.98	5.16
<i>NR</i> (%)	1.63	1.57	0.40	0.96	3.27
<i>NI</i> (%)	1.09	1.07	0.62	0.21	3.82
<i>NO</i> (%)	1.14	1.08	0.67	0.24	5.04
<i>NSR</i> (%)	0.65	0.56	0.69	-1.17	3.61
<i>NEIO</i> (%)	-0.04	-0.01	0.33	-2.41	2.23
<i>RET</i> (%)	0.86	1.36	4.49	-22.53	12.85
<i>EXRET</i> (%)	0.47	1.01	4.48	-23.14	12.43

Table 2. Summary statistics.

Read:

- Table 1 illustrates the paper's key measurement idea: exchanges into and out of fund categories can be used to infer shifts between bond-like and equity-like mutual funds.
- Net exchanges sum approximately to zero across all categories, so net exchanges into equity funds correspond to shifts away from bond funds.
- Table 2 shows that average normalized net exchanges are close to zero, while net sales are positive on average. This supports the view that *NEIO* captures reallocations rather than secular growth in mutual fund assets.

Economic message: the paper's contribution begins with measurement. The authors isolate a flow component that looks more like active investor sentiment than ordinary savings flows.

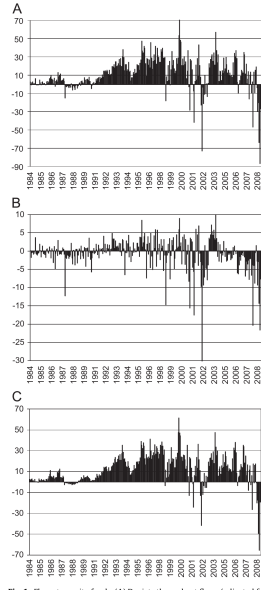


Fig. 1. Flows to equity funds. (A) Depicts the real net flows (adjusted for inflation and presented in December 2008 prices) to the equity funds from February 1984 to December 2008, a period of 299 months. The real net flows (in billions \$) are composed of "new sales" plus "exchanges in" minus "redemptions" and "exchanges out." (B) Depicts the real net exchanges (adjusted for inflation and presented in December 2008 prices) to the equity funds from February 1984 to December 2008, a period of 299 months. The real net exchanges (in billions \$) are composed of "exchanges in" minus "exchanges out." (C) Depicts the real net sales (adjusted for inflation and presented in December 2008 prices) to the equity funds from February 1984 to December 2008, a period of 299 months. The real net sales (in billions \$) are composed of "new sales" minus "redemptions."

Fig. 1. Flows to equity funds.

Table 3

Regressions of excess market returns ($ExRET$) on contemporaneous variables.

The table presents the coefficients from the time-series regressions of excess market returns ($ExRET$) on the respective variables. The sample period ranges from February 1984 to December 2008, covering a total of 299 months. $ExRET$ is the monthly excess market return from the CRSP value-weighted index composed of NYSE, Amex, and Nasdaq stocks minus the 30-day T-bill (in %). The net flows of the equity funds and their components are normalized each month by the previous month's fund assets value. $NFLOWS$ is the normalized net flows (in %). $NEIO$ is the normalized "net exchanges" ("exchanges in" minus "exchanges out") in %. NSR is the normalized "net sales" ("new sales" minus "redemptions") in %. $ENEIO$ is $NEIO$'s expected value using a regression of $NEIO$ on four lags of $NEIO$, $ExRET$, and NSR from period $t-1$ to $t-4$. $UENEIO$ is $NEIO$'s unexpected part from that regression. The table presents the p -values of the Wald statistic, for the significance of the regressions. The t -statistics are reported below the coefficients. Both the Wald statistics and the t -statistics are corrected for heteroskedasticity using the White (1980) correction

	(1)	(2)	(3)	(4)	(5)	(6)
INTERCEPT	-1.02 [2.73]	0.91 [4.76]	-0.80 [1.79]	0.59 [1.61]	0.50 [2.42]	0.53 [2.45]
$NFLOWS$	2.46 [6.51]					
$NEIO$		9.85 [10.42]		9.40 [10.67]		
NSR			1.94 [4.17]	0.47 [1.37]		
$ENEIO$						0.85 [0.35]
$UENEIO$					10.88 [10.09]	10.87 [10.07]
Adjusted- R^2	0.223	0.373	0.086	0.375	0.385	0.384
p -Values of Wald statistic for Regression	0.000	0.000	0.000	0.000	0.000	0.000

Table 3. Regressions of excess market returns on contemporaneous variables.

Read:

- Fig. 1 shows that real net flows and net sales trend upward over time, while net exchanges are more stationary and more episodic.
- Table 3 shows that $NEIO$ is strongly contemporaneously related to excess market returns. The adjusted R^2 is about 0.373 using $NEIO$, compared with only 0.086 using NSR .
- When $NEIO$ and NSR are included together, $NEIO$ remains highly significant, while NSR becomes weak and insignificant.

Economic message: the contemporaneous stock-flow relation is primarily about exchanges, not ordinary fund sales and redemptions.

6.2 Return reversal and dynamic price response

Table 4
Regressions of excess market returns ($ExRET$) on lagged variables.

The table presents the coefficients from the time-series regressions of excess market returns ($ExRET$) on the respective variables. The sample period ranges from February 1984 to December 2008, covering a total of 299 months. $ExRET$ is the monthly excess market return from the CRSP value-weighted index composed of NYSE, Amex, and Nasdaq stocks minus the 30-day T-bill return (in %). The flow components of the equity funds are normalized each month by the previous month's fund assets value. $NEIO$ is the normalized "net exchanges" ("exchanges in" minus "exchanges out") in %; $NEIO$ is winsorized around the 1% tails of the distribution. NSR is the normalized "net sales" ("new sales" minus "redemptions") in %. $SumNEIO_{2,4}$ is the sum of $NEIO$ lags from period $t-2$ to $t-4$. $SumNEIO_{1,4}$ is the sum of $NEIO$ lags from period $t-1$ to $t-4$. $UENEIO$ is $NEIO$'s unexpected part from a regression of $NEIO$ on four lags of $NEIO$, $ExRET$, and NSR from period $t-1$ to $t-4$. $SumUENEIO_{2,4}$ is the sum of $UENEIO$ lags from period $t-2$ to $t-4$. The table presents the p -values of the Wald statistic for the significance of the regressions and the variables. For example, the p -value for $NEIO$ lags is the p -value of the Wald statistic for the significance of the $NEIO$ lags as a group. The t -statistics are reported below the coefficients. Both the Wald statistics and the t -statistics are corrected for heteroskedasticity using the White (1980) correction.

	(1)	(2)	(3)	(4)	(5)	(6)
INTERCEPT	0.332 [1.29]	0.343 [1.31]	0.333 [1.30]	0.395 [0.99]	-0.426 [0.92]	0.498 [1.92]
$NEIO(t-1)$	-0.338 [0.39]				-1.768 [2.05]	
$NEIO(t-2)$	-0.940 [1.10]				-1.506 [1.52]	
$NEIO(t-3)$	-1.289 [1.48]				-1.254 [1.10]	
$NEIO(t-4)$	-1.844 [2.21]				-2.455 [2.68]	
$SumNEIO_{2,4}$		-1.392 [3.10]				
$SumUENEIO_{2,4}$						-1.695 [3.27]
$SumNEIO_{1,4}$			-1.113 [3.07]			
$NSR(t-1)$				1.035 [1.50]	1.695 [2.28]	
$NSR(t-2)$				-0.353 [0.44]	-0.026 [0.03]	
$NSR(t-3)$				-1.162 [1.66]	-1.454 [1.62]	
$NSR(t-4)$				0.673 [1.06]	0.806 [1.23]	
Adjusted- R^2	0.013	0.021	0.019	0.000	0.031	0.027
p -Values of Wald statistic for Regression	0.026	0.002	0.002	0.274	0.003	0.001
$NEIO$ lags	0.026			0.274	0.001	
NSR lags					0.021	

Table 4. Regressions of excess market returns on lagged variables.

Read:

- Table 4 is the central reversal evidence. Lagged $NEIO$ predicts negative future market excess returns.
- The sum of lags $t-2$ through $t-4$ is significantly negative, which means public information about past net exchanges can forecast subsequent reversal.
- Table 5 summarizes the VAR impulse response. A one-standard-deviation $NEIO$ shock is associated with 1.95% contemporaneous excess return and about full reversal by ten months.

Economic message: the market reaction to net exchanges is not permanent. This is why the authors interpret it as price pressure or noise rather than fundamental information.

Table 5

Impulse response simulation of $ExRET$, $NEIO$, and NSR .

The table presents the impulse response function results of a one-standard-deviation shock to $NEIO$ (normalized net exchanges) on $ExRET$ (the excess market return) based on two alternative estimations: "Full VAR" is the impulse response estimation based on a three-variable VAR system of $NEIO$, NSR , and $ExRET$ with four lags of each variable (see Eq. (1)). "Restricted VAR" is the impulse response estimation based on a three-variable VAR system of $NEIO$, NSR , and $ExRET$ without the lags of $ExRET$ (see Eq. (2)). Period t is the contemporaneous period of the shock. The accumulated response column presents the accumulated periodic effect. The table presents the t -statistics of the impulse response estimation for the accumulated response from period $t+1$ to $t+10$. The t -statistics are calculated based on simulated standard deviations.

Period	Accumulated response	
	Full VAR	Restricted VAR
t	1.95%	1.95%
$t+1$	1.52%	1.54%
$t+2$	1.14%	1.16%
$t+3$	0.90%	0.90%
$t+4$	0.31%	0.28%
$t+5$	0.16%	0.12%
$t+6$	0.17%	0.13%
$t+7$	0.08%	0.03%
$t+8$	0.03%	-0.01%
$t+9$	0.03%	-0.01%
$t+10$	-0.01%	-0.04%
t -Statistic	3.03	3.46
% of reversal	101	102

Table 5. Impulse response simulation of $ExRET$, $NEIO$, and NSR .

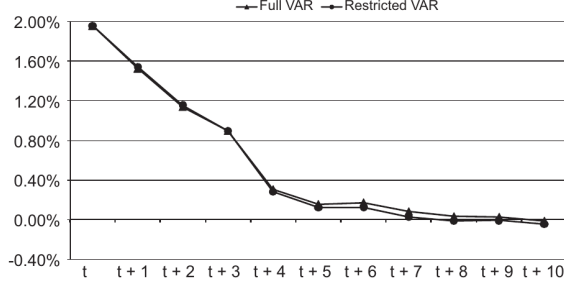


Fig. 2. Accumulated impulse response function. Fig. 2 presents the accumulated impulse response simulation of a one-standard-deviation shock to *NEIO* (normalized net exchanges) on *ExRET* (excess market return) based on two alternative estimations: “Full VAR” is the impulse response estimation based on a three-variable VAR system of *NEIO*, *NSR*, and *ExRET* with four lags of each variable (see Eq. (1)); “Restricted VAR” is the impulse response estimation based on a three-variable VAR system of *NEIO*, *NSR*, and *ExRET* without the lags of *ExRET* (see Eq. (2)). Period *t* is the contemporaneous period of the shock

Fig. 2. Accumulated impulse response function.

Read:

- Fig. 2 visualizes the reversal: the cumulative response falls steadily after the initial positive return and is essentially gone by month ten.
- Table 6 shows the in-sample strategy. In months when the signal says to be in the market, average excess returns are 1.10%; in the other months, average excess returns are -0.19%.
- The standard deviations of the two return series are similar, so the return difference is not just a volatility difference.

Economic message: the reversal is large enough to matter economically, not just statistically.

Table 6

A trading strategy based on normalized net exchanges.

The table presents the results of a strategy based on the normalized net exchanges (*NEIO*). For each month *t*, we sum the lags of *NEIO* from month *t*−2 to *t*−4 (*SumNEIO*_{*t*−4}), to produce an indicator for month *t* return (recall that ICI data for month *t*−1 are available only at the end of month *t*). If the indicator is negative, the investment decision for month *t* is to invest in the stock market. If the indicator is positive, the investment decision is to invest in a risk-free asset (30-day T-bills). The table presents, for each series of months (“in the stock market,” “out of the stock market”), the average and standard deviation of market return (*RET*) and market excess return over the risk-free rate (*ExRET*).

Decision proportions			Average		Std	
Decision	Obs	%	<i>RET</i> (%)	<i>ExRET</i> (%)	<i>RET</i> (%)	<i>ExRET</i> (%)
In	161	54.6	1.50	1.10	4.63	4.60
Out	134	45.4	0.17	−0.19	4.24	4.26

Table 6. A trading strategy based on normalized net exchanges.

6.3 Trading strategy and robustness

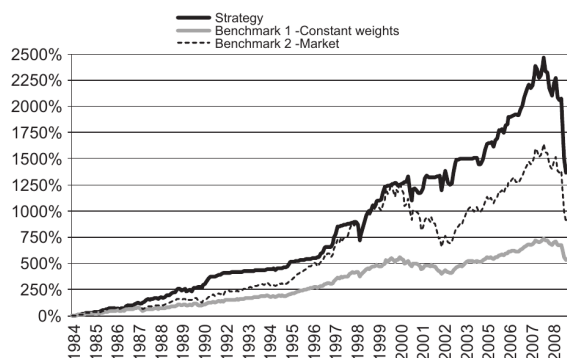


Fig. 3. Accumulated return of the in-sample trading strategy. Fig. 3 depicts the accumulated returns (from June 1984 to December 2008) of three investments: Strategy—investment based on the indicator decision (the solid black line); Benchmark 1—investment in constant weights based on the indicator percentage of months in (54.6%) and out (45.4%) of the market (the solid gray line); Benchmark 2—investment 100% in the market portfolio (the dashed black line).

Fig. 3. Accumulated return of the in-sample trading strategy.

Read:

- Fig. 3 shows that the in-sample switching strategy outperforms both a constant-weight benchmark and the market over the sample.
- Table 7 repeats the exercise out-of-sample using a rolling 36-month window. The market months again have higher average returns than the non-market months.
- These results are meant to demonstrate economic significance; they are not presented as a risk-free arbitrage.

Economic message: publicly observable mutual fund exchange data contain useful information about future market reversals.

Table 7

An out-of-sample trading strategy based on *NEIO*.

The table presents the results of an out-of-sample strategy based on the normalized net exchanges (*NEIO*). For each month t , we look at the previous 36 months and sum the lags of *NEIO* from month $t-2$ to $t-4$ ($SumNEIO_{2-4}$), to produce an indicator for the month t return (recall that ICI data for month $t-1$ are available only at the end of month t). If the correlation of the indicator with *ExRET* (the excess market return over the 30-day T-bill return) is negative and the indicator at month t is positive (negative), the investment decision for month t is to invest in the risk-free rate asset (stock market). If the correlation of the indicator with *ExRET* is positive and the indicator at month t is positive (negative), the investment decision for month t is to invest in the stock market (risk-free rate). The table presents, for each series of months ("in the stock market," "out of the stock market"), the average and standard deviation of market return (*RET*) and market excess return over the risk-free rate (*ExRET*).

Decision proportions			Average		Std	
Decision	Obs	%	<i>RET</i> (%)	<i>ExRET</i> (%)	<i>RET</i> (%)	<i>ExRET</i> (%)
In	133	50.6	1.37	0.99	4.55	4.53
Out	130	49.4	0.08	-0.24	4.31	4.32

Table 7. An out-of-sample trading strategy based on *NEIO*.

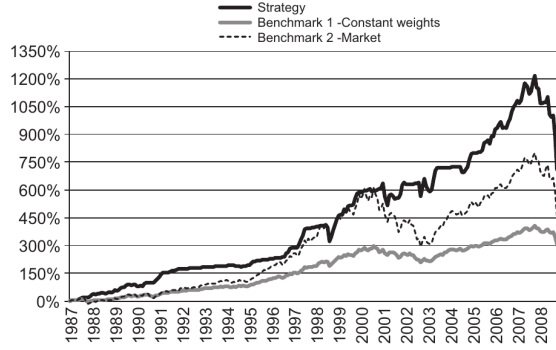


Fig. 4. Accumulated return of the out-of-sample trading strategy. Depicts the accumulated returns (from February 1987 to December 2008) of three investments: Strategy—investment based on the indicator decision (the solid black line); Benchmark 1—investment in constant weights based on the indicator percentage of months in (50.6% in and 49.4% out) of the market (the solid gray line); Benchmark 2—investment 100% in the market portfolio (the dashed black line).

Fig. 4. Accumulated return of the out-of-sample trading strategy.

Read:

- Fig. 4 shows that the out-of-sample strategy also outperforms the market and constant-weight benchmark over the rolling strategy period.
- Fig. 5 shows that the impulse-response pattern is present in both subperiods, though the exact timing and magnitude differ.
- The robustness section also checks January effects, winsorization, raw market returns, exclusion of mixed funds, and alternate flow proxies.

Economic message: the reversal is not concentrated in a single narrow episode. It persists across subsamples and robustness choices.

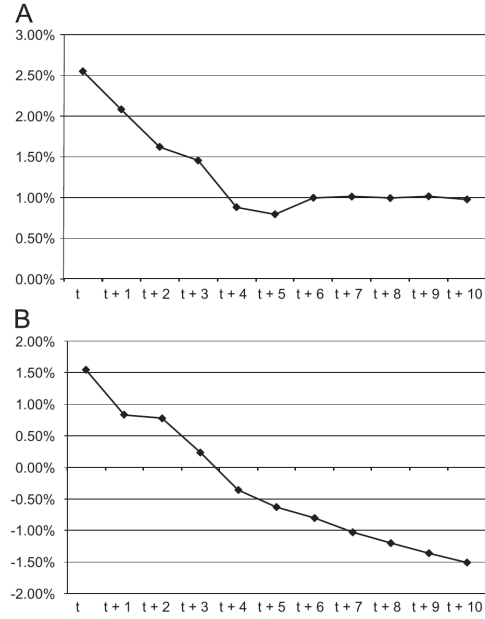


Fig. 5. Accumulated impulse response function—two subperiods. (A) and (B) depict the accumulated impulse response simulation of a one-standard-deviation shock to *NEIO* (normalized net exchanges) on *ExRET* (excess market return) for the two subperiods. The impulse response

Fig. 5. Accumulated impulse response function - two subperiods.

6.4 Links to VIX, rates, and sentiment measures

Table 8
Regressions of *NEIO* on selected explanatory variables.

The table presents the coefficients from the time-series regressions of *NEIO* as a dependent variable. For each explanatory variable: the “Sample year begin” is the first year for which we have data about the explanatory variable; the “Sample year end” is the last year for which we have data about the explanatory variable; *N* is the number of available monthly observations. *NEIO* is the normalized “net exchanges” (“exchanges in” minus “exchanges out”) in %. The measure is winsorized around the 1% tails of the distribution. $RF(t-1)$ is the risk-free rate for 30 days based on the 30-day T-bill return from CRSP at the end of the month (assuming the return in month t reflects the 30-day interest rate known to investors at the end of the previous month). $RFdiff$ is the contemporaneous change in RF (the return difference of $RF_{t-1} - RF_t$). $VIX(t-1)$ is the volatility index of Chicago Board Options Exchange (CBOE) retrieved from finance.yahoo.com (symbol ‘vix’) at the end of the previous month. For convenience, the measure, which is in percentage terms, is divided by 100 (for example, 20 is converted into 0.2). $VIXdiff$ is the contemporaneous change in VIX . $CSI(t-1)$ is the University of Michigan “Consumer Sentiment Index” at the end of the previous month. $CSdiff$ is the contemporaneous change in CSI . $BW(t-1)$ is the Baker and Wurgler (2006) orthogonal sentiment measure (SENT⁺) at the end of the previous month. $BWdiff$ is the contemporaneous change in BW . The t -statistics are reported below the coefficients and corrected for heteroskedasticity using the White (1980) correction. Due to the persistence of the explanatory variables, the table also presents the Amihud and Hurvich (2004) correction for a single persistent regressor (A-H). Simulated p -values (one-sided sim p -value) are computed as in Boudoukh, Michaely, Richardson, and Roberts (2007, Table II), via 10,000 simulations under the null of zero predictability, but accounting for the regressor’s autocorrelation and the cross-correlation of the errors.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
INTERCEPT	0.110 [4.55]	0.003 [0.24]	0.075 [2.18]	-0.043 [0.04]	0.236 [1.88]	-0.040 [2.37]	-0.039 [2.20]	-0.041 [2.25]
$VIX(t-1)$	-0.354 [4.82]							
$VIXdiff$		-1.708 [4.71]						
$RF(t-1)$			-0.308 [2.88]					
$RFdiff$				0.463 [1.06]				
$CSI(t-1)$					-0.003 [2.20]			
$CSdiff$						0.013 [2.91]		
$BW(t-1)$							-0.019 [0.69]	
$BWdiff$								0.027 [0.51]
R^2	0.054	0.157	0.039	0.008	0.009	0.030	0.002	0.001
Sample year begin	1990	1990	1984	1984	1984	1984	1984	1984
Sample year end	2008	2008	2008	2008	2007	2007	2005	2005
<i>N</i>	227	227	299	299	280	280	264	264
A-H correction								
Regressor persistence	0.86		0.96		0.91		0.84	
Residual correlation	-0.48		0.06		0.16		0.02	
New coefficient	-0.59		-0.303		0.00		-0.02	
New t -statistic	-4.33		-2.87		-2.08		-0.68	
One-sided sim p -value	0.000		0.003		0.023		0.245	

Table 8. Regressions of *NEIO* on selected explanatory variables.

Table 9
Regressions of excess market return on *NEIO*’s explanatory variables.

The table presents the coefficients from the time-series regressions of the excess market return ($ExRET$) as the dependent variable. Due to data limitations, the regression sample period depends on the explanatory variables. For each explanatory variable: the “Sample year begin” is the first year for which we have data about the explanatory variable; the “Sample year end” is the last year for which we have data about the explanatory variable; *N* is the number of available monthly observations. $ExRET$ is the monthly excess market return from the CRSP value-weighted index composed of NYSE, Amex, and Nasdaq stocks minus the 30-day T-bill (in %). $RF(t-1)$ is the risk-free rate for 30 days based on the 30-day T-bill return from CRSP at the end of the month (assuming the return in month t reflects the 30-day interest rate known to investors at the end of the previous month). $RFdiff$ is the contemporaneous change in RF (the return difference of $RF_{t-1} - RF_t$). $VIX(t-1)$ is the volatility index of CBOE retrieved from finance.yahoo.com (symbol ‘vix’) at the end of the previous month. For convenience, the measure, which is in percentage terms, is divided by 100 (for example, 20 is transformed to 0.2). $VIXdiff$ is the contemporaneous change in VIX . $CSI(t-1)$ is the University of Michigan “Consumer Sentiment Index” at the end of the previous month. $CSdiff$ is the contemporaneous change in CSI . $BW(t-1)$ is the Baker and Wurgler (2006) orthogonal sentiment measure (SENT⁺) at the end of the previous month. $BWdiff$ is the contemporaneous change in BW . The t -statistics are reported below the coefficients and corrected for heteroskedasticity using the White (1980) correction. Due to the persistence of the explanatory variables, the table also presents the Amihud and Hurvich (2004) correction for a single persistent regressor (A-H). Simulated p -values (one-sided sim p -value) are computed, as in Boudoukh, Michaely, Richardson, and Roberts (2007, Table II) via 10,000 simulations under the null of zero predictability, but accounting for the regressor’s autocorrelation and the cross-correlation of the errors.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
INTERCEPT	0.947 [0.99]	0.447 [2.14]	0.117 [0.18]	0.496 [1.97]	4.130 [1.59]	0.684 [2.70]	0.739 [2.83]	0.656 [2.43]
$VIX(t-1)$	-2.800 [0.51]							
$VIXdiff$		-72.887 [11.01]						
$RF(t-1)$			0.897 [0.58]					
$RFdiff$				13.101 [1.70]				
$CSI(t-1)$					-0.037 [1.31]			
$CSdiff$						0.222 [3.43]		
$BW(t-1)$							-0.829 [1.84]	
$BWdiff$								-0.296 [0.28]
R^2	0.002	0.472	0.001	0.024	0.006	0.038	0.017	0.001
Sample year begin	1990	1990	1984	1984	1984	1984	1984	1984
Sample year end	2008	2008	2008	2008	2007	2007	2005	2005
<i>N</i>	227	227	299	299	280	280	264	264
A-H correction								
Regressor persistence	0.86		0.96		0.91		0.84	
Residual correlation	-0.72		0.16		0.18		-0.07	
New coefficient	-4.07		1.08		-0.01		-0.83	
New t -statistic	-0.87		0.72		-1.16		-1.91	
One-sided sim p -value	0.224		0.255		0.117		0.036	

Table 9. Regressions of excess market return on *NEIO*’s explanatory variables.

Read:

- Table 8 shows that *NEIO* is strongly negatively related to VIX and changes in VIX. Investors exchange into equity funds when market fear decreases.
- *NEIO* is only weakly related to short rates and the University of Michigan Consumer Sentiment Index, and it is not meaningfully related to Baker–Wurgler sentiment.
- Table 9 shows that the explanatory variables by themselves do not reproduce the key reversal pattern. VIX-related variables matter contemporaneously but do not explain the later reversal.

Economic message: *NEIO* is related to market risk perceptions, especially VIX, but it also captures a separate sentiment or demand-pressure component.

Table 10
Regressions of excess market return on *NEIO* and VIX.

The table presents the coefficients from the time-series regressions of the excess market return (*ExRET*) as the dependent variable. The sample period ranges from June 1990 to December 2008. *ExRET* is the monthly excess market return from the CRSP value-weighted index composed of NYSE, Amex, and Nasdaq stocks minus the 30-day T-bill (in %). *NEIO* is the normalized “net exchanges” (“exchanges in” minus “exchanges out”) in %. *NEIO* is winsorized around the 1% level of the distribution. *NSR* is the normalized “net sales” (“new sales” minus “redemptions”) in %. *SumNEIO_{t-4}* is the sum of *NEIO* lags from period $t-1$ to $t-4$. *VIX* is the volatility index of CBOE retrieved from finance.yahoo.com (symbol “vix”). For convenience, the measure, which is in percentage terms, is divided by 100 (for example, 20 is transformed to 0.2). *VIXdiff* is the contemporaneous change in VIX at period t . *SumVIXdiff_{t-4}* is the sum of *VIXdiff* lags from period $t-1$ to $t-4$. “YES” in “*ExRET* and *NSR* lags” indicates that the regression contains four lags of *ExRET* and *NSR* from period $t-1$ to $t-4$. The t -statistics are reported below the coefficients and corrected for heteroskedasticity using the White (1980) correction. Due to the persistence of the explanatory variables, the table also presents the Amihud and Hurvich (2004) correction for a single persistent regressor (A-H) and the Amihud, Hurvich, and Wang (2008) correction for multiple persistent regressors (A-H-W). Simulated p -values (one-sided sim p -value) are computed, as in Boudoukh, Michaely, Richardson, and Roberts (2007, Table II), via 10,000 simulations under the null of zero predictability, but accounting for the regressors’ auto-correlation and cross-correlations and the cross-correlation of the errors.

	(1)	(2)	(3)	(4)	(5)
<i>INTERCEPT</i>	0.45 [2.51]	0.39 [1.40]	0.38 [1.33]	-0.57 [0.88]	-0.60 [0.93]
<i>NEIO</i>	10.26 [6.21]				
<i>VIXdiff</i>	-55.84 [9.41]				
<i>VIXdiff(t-1)</i>		-18.94 [1.47]			
<i>VIXdiff(t-2)</i>		6.09 [0.80]			
<i>VIXdiff(t-3)</i>		6.21 [0.80]			
<i>VIXdiff(t-4)</i>		8.69 [0.92]			
<i>SumVIXdiff_{t-4}</i>			-0.23 [0.03]		3.45 [0.42]
<i>SumNEIO_{t-4}</i>				-2.55 Yes [2.19]	-2.51 [2.15]
<i>ExRET</i> and <i>NSR</i> lags		No	No	Yes	Yes
Adjusted- R^2	0.606	0.028	-0.005	0.012	0.009
A-H and A-H-W corrections					
New VIX coefficient			-0.96		3.10
New t -statistic			-0.19		0.44
One-sided sim p -value			0.445		0.315
New <i>NEIO</i> coefficient				-2.35	-2.43
New t -statistic				-1.98	-2.07
One-sided sim p -value				0.023	0.019

Table 10. Regressions of excess market return on *NEIO* and VIX.

Read:

- Table 10 shows that *NEIO* remains important when VIX changes are included.
- Contemporaneous changes in VIX explain part of current market returns, but lagged VIX changes do not remove the negative predictive power of lagged *NEIO*.
- Table 11 shows that adding Michigan or Baker–Wurgler sentiment variables does not eliminate the *NEIO* result.

Economic message: the authors’ sentiment proxy is not simply a relabeling of standard sentiment or fear measures.

6.5 Cross-sectional evidence: small stocks and growth stocks

Table 12
Regressions of size portfolio returns.

The table presents the coefficients from the time-series regressions of small and big size cap portfolios. Panel A reports the results of Fama-French SMB portfolios on *NEIO* and *SumNEIO_{t-4}*. Panel B reports the results of Russell 1000 and Russell 2000 total return indexes on *NEIO* and *SumNEIO_{t-4}*. The sample period ranges from February 1984 to December 2008, covering a total of 299 months. In each panel, specifications (1) and (2) are for the small portfolio, specifications (3) and (4) are for the big size portfolio, and specifications (5) and (6) are for the “small-minus-big” (SMB) portfolio. *NEIO* is the normalized “net exchanges” (“exchanges in” minus “exchanges out”) in %. *NEIO* is winsorized around the 1% tails of the distribution. *SumNEIO_{t-4}* is the sum of *NEIO* lags from period $t-2$ to $t-4$. The t -statistics are reported below the coefficients and are corrected for heteroskedasticity using the White (1980) correction.

	Small		Big		SMB	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A—FF’s SMB factor						
<i>INTERCEPT</i>	1.46 [5.95]	0.82 [2.49]	1.36 [7.41]	0.80 [3.23]	0.10 [0.52]	0.01 [0.07]
<i>NEIO</i>	11.03 [8.80]		9.42 [10.31]		1.61 [2.58]	
<i>SumNEIO_{t-4}</i>		-1.57 [2.91]		-1.37 [3.12]		-0.20 [0.63]
Adjusted- R^2	0.314	0.017	0.374	0.022	0.015	0.000
	Russell 2000		Russell 1000		Rus2-Rus1	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel B—Russell 1000 and Russell 2000						
<i>INTERCEPT</i>	1.30 [5.31]	0.62 [1.88]	1.30 [6.78]	0.72 [2.82]	0.00 [0.01]	-0.10 [0.34]
<i>NEIO</i>	11.41 [8.85]		9.61 [10.31]		1.80 [2.73]	
<i>SumNEIO_{t-4}</i>		-1.79 [3.11]		-1.48 [3.29]		-0.31 [0.96]
Adjusted- R^2	0.326	0.022	0.362	0.024	0.023	0.000

Table 12. Regressions of size portfolio returns.

Read:

Table 11
Regressions of excess market return on *NEIO* and sentiment measures.

The table presents the coefficients from the time-series regressions of the excess market return (*ExRET*) as the dependent variable. The sample period ranges from June 1986 to December 2005. *ExRET* is the monthly excess market return from the CRSP value-weighted index composed of NYSE, Amex, and Nasdaq stocks minus the 30-day T-bill (in %). *NEIO* is the normalized “net exchanges” (“exchanges in” minus “exchanges out”) in %. *NEIO* is winsorized around the 1% level of the distribution. *SumNEIO_{t-4}* is the sum of *NEIO* lags from period $t-1$ to $t-4$. *CSI* is the University of Michigan “Consumer Sentiment Index” at the end of the month. *CSdiff* is the contemporaneous change in *CSI*. *SumCSdiff_{t-4}* is the sum of *CSdiff* lags from period $t-1$ to $t-4$. *BW(t-1)* is the Baker and Wurgler (2006) orthogonal sentiment measure (*SENT*⁺) at the end of the previous month. *BWdiff* is the contemporaneous change in *BW*. *SumBWdiff_{t-4}* is the sum of *BWdiff* lags from period $t-1$ to $t-4$. Specifications (5)–(7) include as controls the four lags of *ExRET* and *NSR* from period $t-1$ to $t-4$. The t -statistics are reported below the coefficients and corrected for heteroskedasticity using the White (1980) correction. Due to the persistence of the explanatory variables, the table also presents the Amihud and Hurvich (2004) correction for a single persistent regressor (A-H), and the Amihud, Hurvich, and Wang (2008) correction for multiple persistent regressors (A-H-W). Simulated p -values (one-sided sim p -values) are computed, as in Boudoukh, Michaely, Richardson, and Roberts (2007, Table II), via 10,000 simulations under the null of zero predictability, but accounting for the regressors’ auto-correlation and cross-correlations and the cross-correlation of the errors.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>INTERCEPT</i>	1.04 [5.08]	1.03 [5.06]	1.03 [5.09]	1.03 [5.06]	0.12 [0.22]	0.11 [0.21]	0.09 [0.16]
<i>NEIO</i>	9.58 [9.95]	9.37 [9.87]	9.61 [9.99]	9.39 [9.97]			
<i>CSdiff</i>		0.09 [1.77]		0.09 [1.71]			
<i>BWdiff</i>			-0.51 [0.67]	-0.45 [0.59]			
<i>SumNEIO_{t-4}</i>					-1.41 [2.77]	-1.27 [2.55]	-1.30 [2.50]
<i>SumCSdiff_{t-4}</i>						-0.03 [0.68]	-0.03 [0.61]
<i>BW(t-1)</i>						-0.62 [1.20]	
<i>SumBWdiff_{t-4}</i>							-0.46 [0.76]
Adjusted- R^2	0.390	0.394	0.390	0.392	0.027	0.028	0.023
A-H and A-H-W corrections							
New <i>NEIO</i> coefficient					-1.32	-1.17	-1.22
New t -statistic					-2.44	-2.29	-2.30
One-sided sim p -value					0.004	0.029	0.021
New <i>CSI</i> coefficient						-0.03	-0.03
New t -statistic						-0.72	-0.64
One-sided sim p -value						0.288	0.307
New <i>BW</i> coefficient						-0.63	-0.47
New t -statistic						-1.35	-0.79
One-sided sim p -value						0.094	0.203

Table 11. Regressions of excess market return on *NEIO* and sentiment measures.

Table 13
Regressions of value and growth portfolio returns.

The table presents the coefficients from the time-series regressions of Fama-French HML portfolios. The sample period ranges from February 1984 to December 2008, covering a total of 299 months. Specifications (1) and (2) are for the high book-to-market (value) portfolio, specifications (3) and (4) are for the low book-to-market (growth) portfolio, and specifications (5) and (6) are for the “high-minus-low” portfolio. *NEIO* is the normalized “net exchanges” (“exchanges in” minus “exchanges out”) in %. *NEIO* is winsorized around the 1% tails of the distribution. *SumNEIO_{t-4}* is the sum of *NEIO* lags from period $t-2$ to $t-4$. The t -statistics are reported below the coefficients and are corrected for heteroskedasticity using the White (1980) correction.

	High		Low		HML	
	(1)	(2)	(3)	(4)	(5)	(6)
FF’s HML factor						
<i>INTERCEPT</i>	1.50 [7.93]	0.98 [3.74]	1.27 [5.15]	0.55 [1.68]	0.22 [1.27]	0.43 [2.35]
<i>NEIO</i>	9.28 [8.71]		11.65 [10.21]		-2.37 [5.02]	
<i>SumNEIO_{t-4}</i>		-1.06 [2.44]		-2.04 [3.67]		0.98 [3.17]
Adjusted- R^2	0.340	0.011	0.340	0.030	0.044	0.022

Table 13. Regressions of value and growth portfolio returns.

- Table 12 shows that the contemporaneous *NEIO* relation is stronger for small stocks than for large stocks. The reversal pattern also matters more for small stocks.
- Table 13 shows that *NEIO* has a larger effect on low book-to-market growth stocks than on high book-to-market value stocks.
- These cross-sectional patterns match the standard idea that sentiment effects are larger in stocks that are harder to value and harder to arbitrage.

Economic message: the same flow measure that predicts aggregate price pressure has stronger effects where sentiment theories predict it should.

6.6 Appendix VAR relations

Table A1

	(1)	(2)	(3)	(4)	(5)
Dependent	<i>NFLOWS</i>	<i>NEIO</i>	<i>NEIO</i>	<i>NSR</i>	<i>NSR</i>
<i>INTERCEPT</i>	-0.039 [0.67]	-0.074 [3.24]	-0.070 [3.02]	0.040 [0.91]	0.037 [0.82]
<i>ExRET</i> ($t-1$)			-0.002 [0.58]		0.006 [0.97]
<i>ExRET</i> ($t-2$)			0.003 [0.71]		-0.003 [0.62]
<i>ExRET</i> ($t-3$)			-0.005 [1.24]		0.004 [0.60]
<i>ExRET</i> ($t-4$)			-0.002 [0.46]		0.000 [0.04]
<i>NEIO</i> ($t-1$)	-0.124 [0.62]	0.002 [0.02]	0.022 [0.25]	-0.119 [0.72]	-0.171 [1.03]
<i>NEIO</i> ($t-2$)	0.074 [0.39]	-0.054 [0.55]	-0.089 [0.80]	0.101 [1.02]	0.143 [1.32]
<i>NEIO</i> ($t-3$)	0.049 [0.24]	0.191 [1.64]	0.234 [1.87]	-0.103 [1.02]	-0.134 [1.10]
<i>NEIO</i> ($t-4$)	-0.180 [0.90]	-0.037 [0.40]	-0.027 [0.26]	-0.118 [0.96]	-0.100 [0.75]
<i>NSR</i> ($t-1$)	0.740 [5.02]	0.086 [1.77]	0.082 [1.77]	0.662 [4.80]	0.649 [4.51]
<i>NSR</i> ($t-2$)	0.105 [0.77]	-0.016 [0.23]	-0.021 [0.29]	0.143 [1.81]	0.153 [1.77]
<i>NSR</i> ($t-3$)	0.059 [0.34]	-0.091 [1.06]	-0.077 [0.85]	0.154 [1.67]	0.143 [1.47]
<i>NSR</i> ($t-4$)	0.074 [0.47]	0.081 [1.30]	0.068 [1.04]	-0.046 [0.47]	-0.034 [0.33]
Adjusted- R^2	0.504	0.043	0.035	0.713	0.710
p -Values of Wald statistic for					
Regression	0.000	0.137	0.352	0.000	0.000
<i>NEIO</i> lags	0.815	0.590	0.449	0.439	0.361
<i>ExRET</i> lags			0.704		0.835
<i>NSR</i> lags	0.000	0.091	0.138	0.000	0.000

Table A1. Main flow relations for the VAR analysis.

Read:

- Table A1 reports the auxiliary flow regressions used in the VAR analysis.
- *NSR* is persistent and helps explain total net flows; *NEIO* is much less predictable.
- The weak predictability of *NEIO* supports the paper's use of *NEIO* innovations and helps explain why contemporaneous *NEIO* and its unexpected component deliver similar results.

7 Key mechanisms and interpretation

7.1 Why net exchanges matter

Net sales and redemptions include household saving, retirement contributions, and withdrawals. Net exchanges are closer to a portfolio allocation decision: investors move money from safer bond-like fund categories into riskier equity-like categories, or vice versa. That makes *NEIO* a better sentiment proxy.

7.2 Why reversal matters

A permanent price effect would be consistent with funds responding to fundamental information. A reversal suggests temporary demand pressure. The paper’s main argument is that sentiment-driven flows push prices away from fundamentals and that later correction produces predictable returns.

7.3 Why VIX is not enough

NEIO is negatively related to VIX and VIX changes. However, VIX-based risk-premium stories cannot generate the magnitude and speed of the observed reversal. The authors therefore interpret *NEIO* as capturing sentiment or demand/supply shocks rather than only changing expected returns.

7.4 Why the effect is stronger for small and growth stocks

Small stocks and growth stocks are harder to value and often harder to arbitrage. Sentiment models therefore predict larger sentiment-driven price pressure in those stocks. The paper finds exactly this pattern.

8 Limitations

- **NEIO is an imperfect measure of bond-equity reallocations.** It only captures exchanges within fund families. Transfers across fund families appear as sales and redemptions, not exchanges.
- **Monthly data miss faster corrections.** Sentiment-induced price pressure that reverses within days or weeks would not be captured well.
- **The strategy is risky, not arbitrage.** The trading strategies have substantial market exposure and are meant to illustrate economic significance.
- **Causality is difficult.** Net exchanges may proxy for a broader set of demand shocks, investor beliefs, or macro risk perceptions.
- **VIX and sentiment are intertwined.** The paper shows VIX cannot explain the reversal, but fear, sentiment, and demand pressure are economically connected.
- **External validity can vary.** The measure depends on the structure and reporting of U.S. mutual fund flows during 1984–2008.

9 Short summary

Ben-Rephael, Kandel, and Wohl build a mutual-fund-flow measure of aggregate investor sentiment using normalized net exchanges into equity funds. The paper shows that these exchanges are strongly positively related to contemporaneous stock market returns but predict negative subsequent returns, implying reversal. A one-standard-deviation shock to net exchanges is associated with a 1.95% contemporaneous excess market return, about 85% of which reverses within four months and essentially all of which reverses within ten months. The effect is stronger in small stocks and growth stocks. The authors interpret the results as evidence that investor sentiment induces temporary noise in aggregate market prices.

10 Conclusion

The paper's main lesson is that the composition of mutual fund flows matters. Total equity fund flows mix together savings, withdrawals, and portfolio reallocation. Net exchanges isolate a cleaner sentiment-like shift between bonds and equities. Those shifts temporarily move aggregate stock prices and later reverse. This makes *NEIO* useful both as a sentiment proxy and as evidence that sentiment can create measurable noise even at the aggregate market level.